MIL-E-1/842G
26 September 1978
SUPERSEDING
MIL-E-1/842F
21 November 1973

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 2050W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Gas, tetrode
Base: See figure 1
Mounting position: Any
Weight: 1.5 ounces (nominal)

ABSOLUTE MAXIMUM RATINGS:

Parameter: / Unit:	Ef V	epx v	epy v	Ec2 Vdc	Ec1 Vdc (See note 1)	Rg Meg
Condition A:	6.3 ±10%	1,300	650	-100	-250	10
Condition B:	6.3 ±10%	350	180	-100	-250	1.0
TEST CONDITIONS:	6.3	---	---	----	----	---

ABSOLUTE MAXIMUM RATINGS:

Parameter: / Unit:	Io mAdc (See note 2)	ik a	tk sec(min)	Ehk Vdc	TA °C
Condition A:	100	1.0	10	-100, +25	-75 to 90
Condition B:	200	1.0	10	-100, +25	-75 to 90
TEST CONDITIONS:	---	---	10	----	-75 to 90

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
3201	Critical grid voltage for conduction (1)	Epp = 460 Vac; Rg = 0.1 Meg; Rp = 3,000 ohms; Ec2 = 0; F = 60; connect pin No. 4 to pin No. 8 See note 6	0.65	II	Ec1	-2.9	-4.5	Vdc
3201	Critical grid voltage for conduction (2)	Epp = 460 Vac; Rg = 10 Meg; Rp = 3,000 ohms; Ec2 = 0; F = 60; connect pin No. 4 to pin No. 8 See note 3	0.65	II	Ec1	---	-7.0	Vdc
3201	Critical anode voltage for conduction	Vary Ebb; Ecc1 = 0; Ec2 = 0; Rg = 0.1 Meg; Rp = 980 ohms; connect pin No. 4 to pin No. 8	0.65	II	Ebb	---	25	Vdc
1231	Pulse emission	e = 180 ±9.0 v; prr = 100 ±5; Ra = 15; Rp = 10; Zm = 7.5; t = 3 sec max; calibrating resistor = 5 ohms	0.65	II	etd	---	76	v
	Quality conformance inspection, part 2							
Appendix D, 30(b)	Dimensions	See figure 1	----	--	---	---	---	---
1031	Low frequency vibration	No voltages	----	--	---	---	---	---
1041	Shock	700 G's	----	--	---	---	---	---
	Shock and vibration test end points:							
3201	Grid voltage 1		----	--	Ec1	-2.5	-5.0	Vdc
3201	Grid voltage 2		----	--	Ec1	---	-8.0	Vdc
1231	Pulse emission		----	--	etd	---	85	v
3201	Anode voltage		----	--	Ebb	---	35	Vdc
3241	Heater current		----	--	If	540	660	mA
1336	Heater-cathode leakage	Ehk = +25 Vdc Ehk = -100 Vdc	---- ----	-- --	Ihk Ihk	--- ---	20 20	μAdc μAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3							
	Life-test provisions	Group B; ib = 500 ma; Ib = 80 mAdc; Rg = 0.05Meg; Epp = 460 Vac; F = 60; t = 500 hours See note 4	---	---	---	---	---	---
	Life-test end points:							
3201	Anode voltage	See note 5	---	---	Ebb	---	60	V
1231	Pulse emission		---	---	etd	---	100	v
1031	Variable-frequency vibration	No voltages See note 7	---	---	---	---	---	---
1031	Vibration-fatigue test	See note 7	---	---	---	---	---	---
	Vibration-fatigue test end points:	See note 7						
3201	Grid voltage (1)		---	---	Ecl	-2.5	-5.0	Vdc
3201	Grid voltage (2)		---	---	Ecl	---	-8.0	Vdc
1231	Pulse emission		---	---	etd	---	85	v
3201	Anode voltage		---	---	Ebb	---	35	Vdc
Appendix D, 30(b)	Dimensions	See figure 1 See note 7	---	---	---	---	---	---

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NOTES:

1. Maximum peak grid 1 to anode voltage with grid negative and anode positive = 750 V.
2. Averaged over any interval of 30 seconds maximum.
3. Use steatite socket (no adapter). Shield the anode supply. Use short shielded anode and grid leads. Anode and grid resistors shall be noninductive. Connect the grid resistor directly at the socket.
4. Phase of grid voltage adjusted to provide start of conduction at peak = ±10 electrical degrees of applied anode voltage.
5. Connect pin No. 4 to pin No. 8. Ebb to vary with Ecl = 0; Rg = 10 Meg; Rp = 980 ohms.
6. This test to be the first test performed at the conclusion of the holding period.
7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test shall be made as a part of Quality Conformance Inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.
8. Reclaimed materials shall be utilized to the maximum extent possible.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 99
DLA - ES

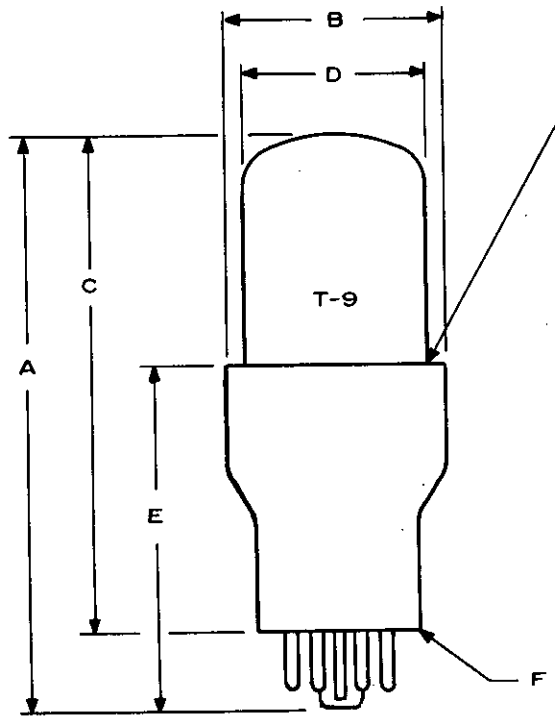
User activities:
Army - AR
Navy - MC, CG
Air Force - 11

Preparing activity:
Air Force - 85

(Project No. 5960-3027)

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PIN CONNECTIONS

PIN NO.	ELEMENT
1	nc
2	h
3	a
4	nc
5	g1
6	g2
7	h
8	k

DIMENSIONS

LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 2				
A		4.13		104.90
C		3.57		90.68

LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSP., PART 3 (NOTE e)				
B		1.57		39.88
D		1.32		33.53
E	2.41	2.47	61.21	62.74
F	BASE: (SEE NOTES b, c and d)			

NOTES:

a. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.

b. DIMENSIONS AND ARRANGEMENT OF BASE PINS AND INDEX GUIDE ARE IDENTICAL WITH THOSE OF JEDEC BASE B8-1.

c. FOR PIN ALINEMENT, USE JEDEC GAGE GB8-1.

d. SPECIAL SKIRTED SMALL SHELL PHENOLIC.

e. SEE NOTE 7.

FIGURE 1. Outline drawing of electron tube type 2050 W

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